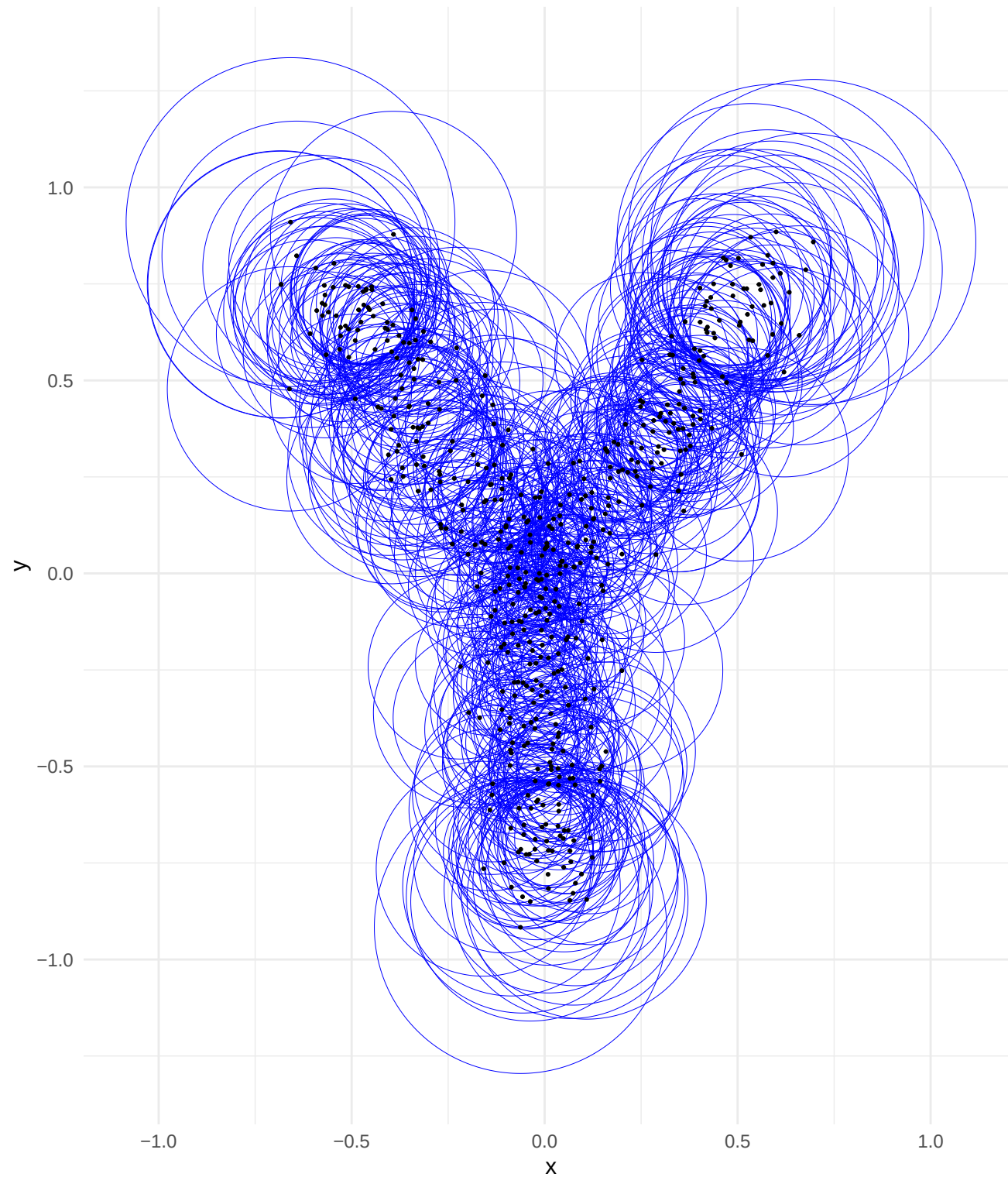
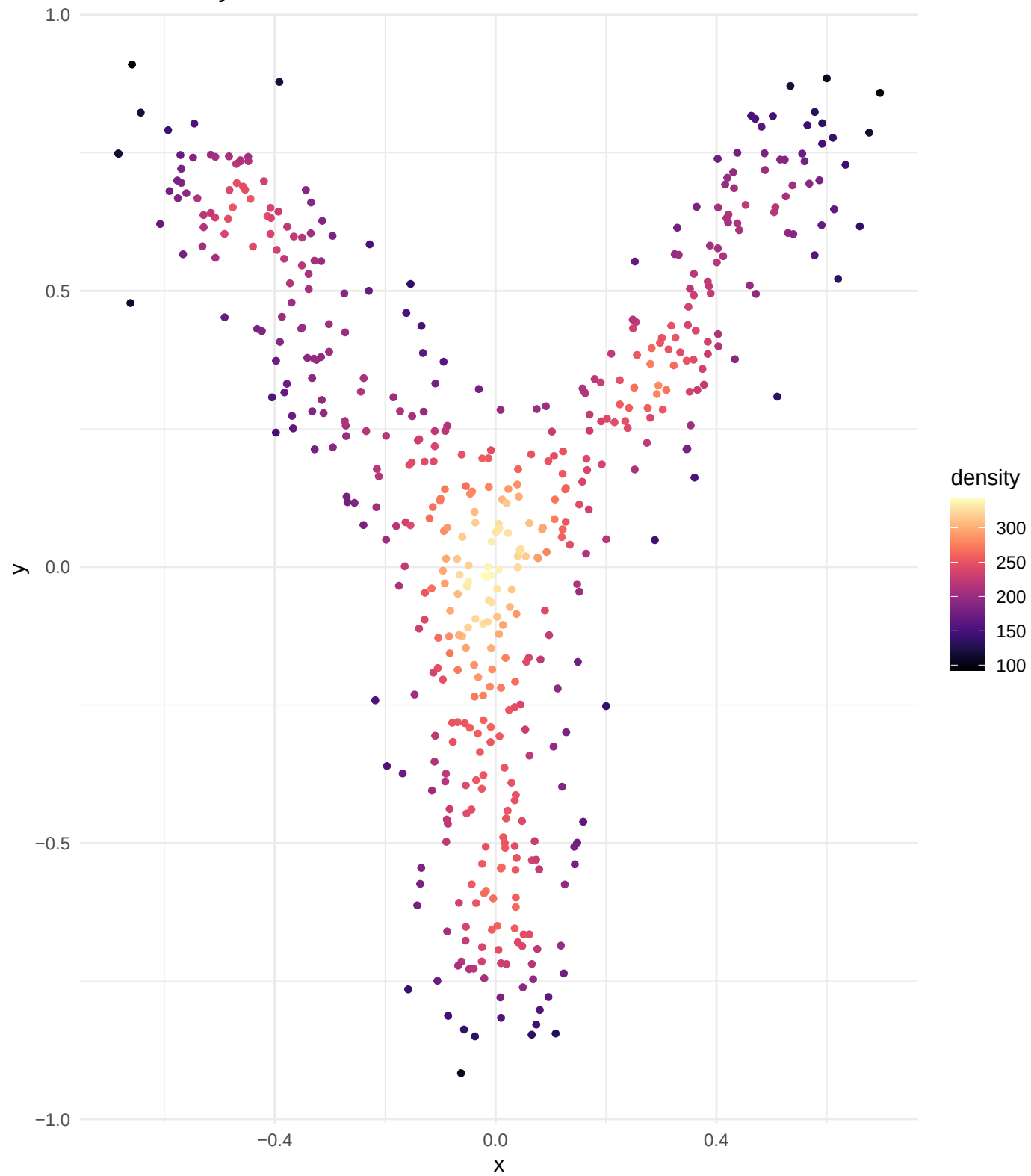


All Adaptive Spheres

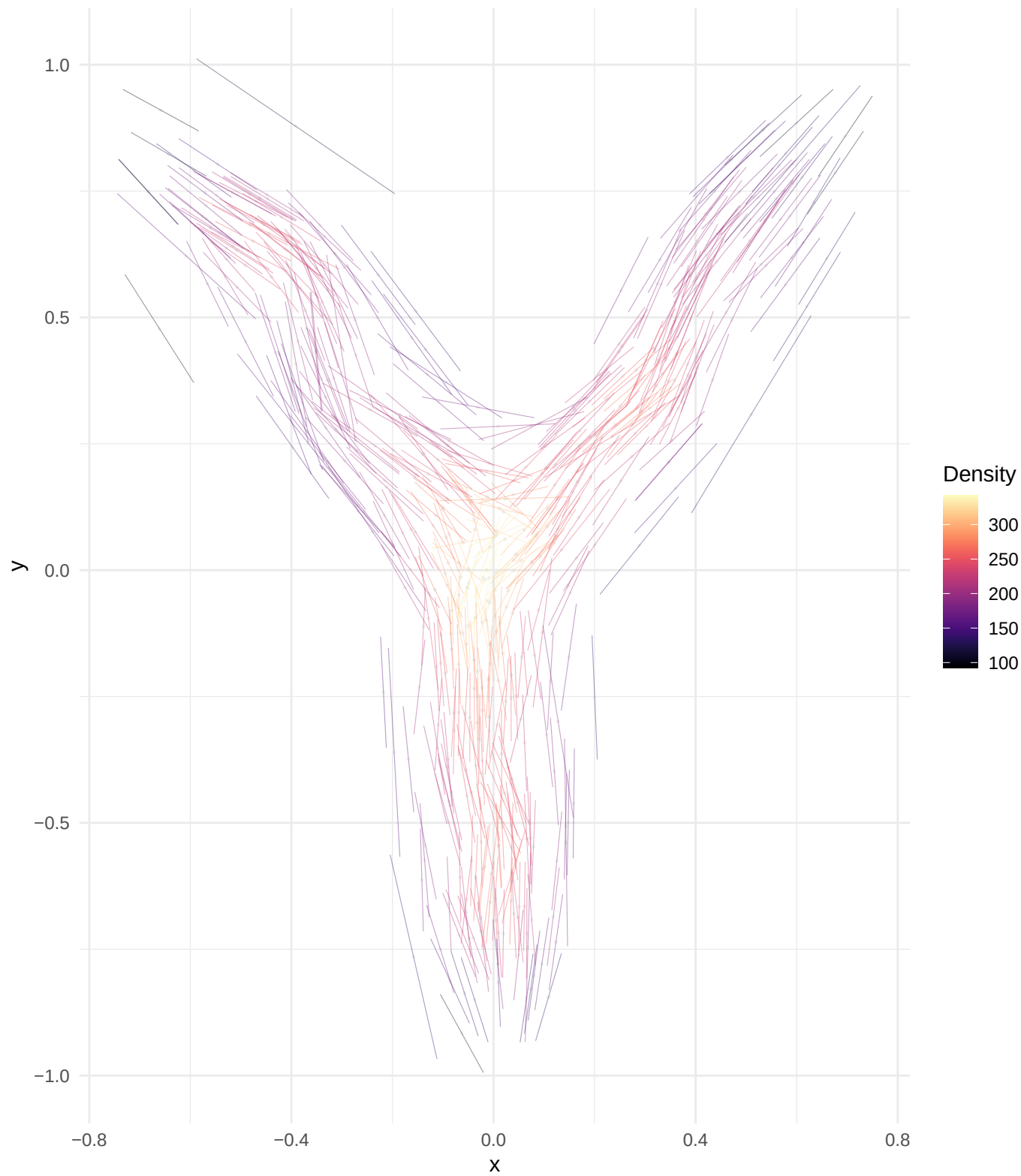
Visualizing local neighborhood size (radius) for every point.



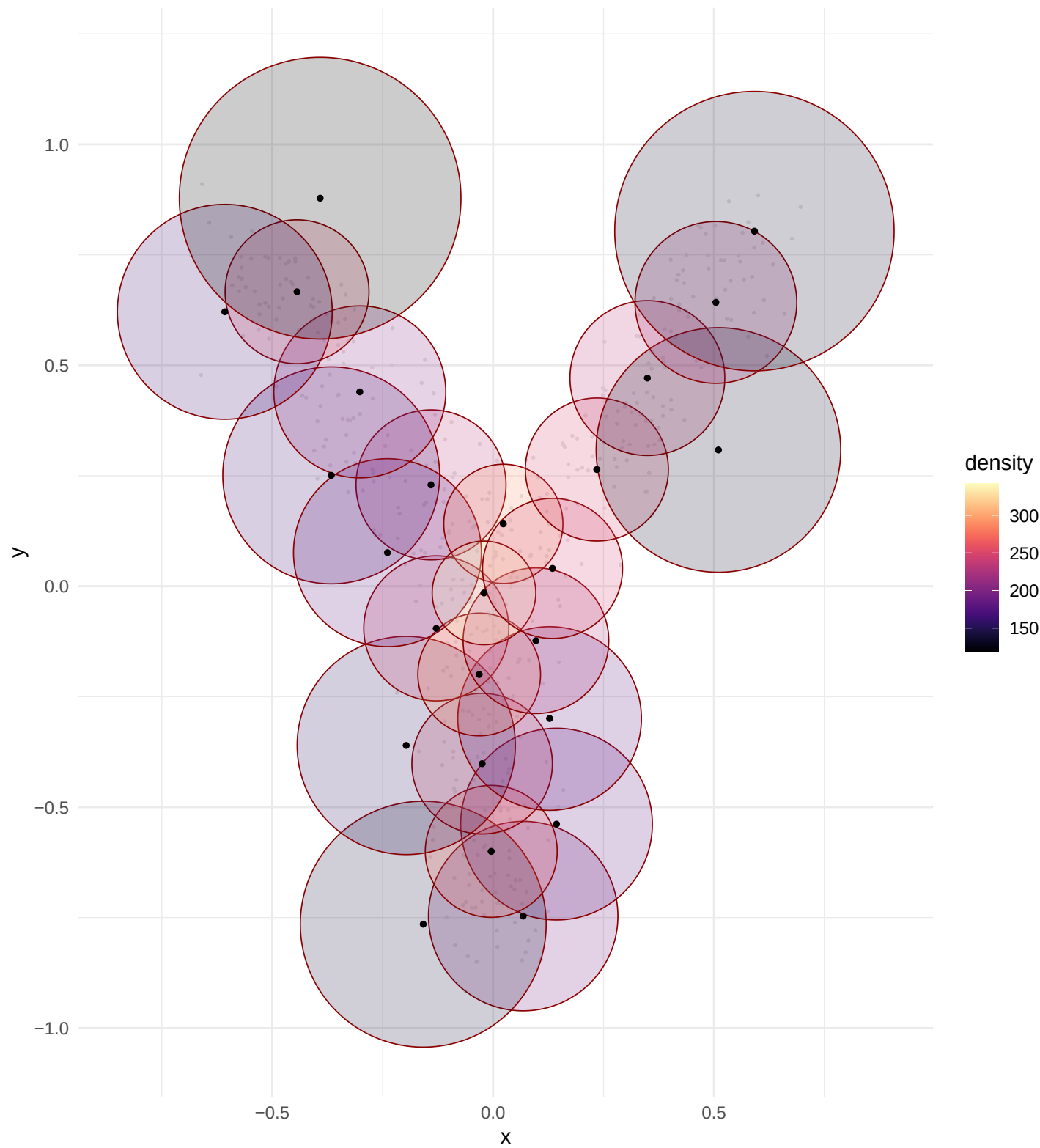
Point Density Distribution



Global Tangent Field (PC1)

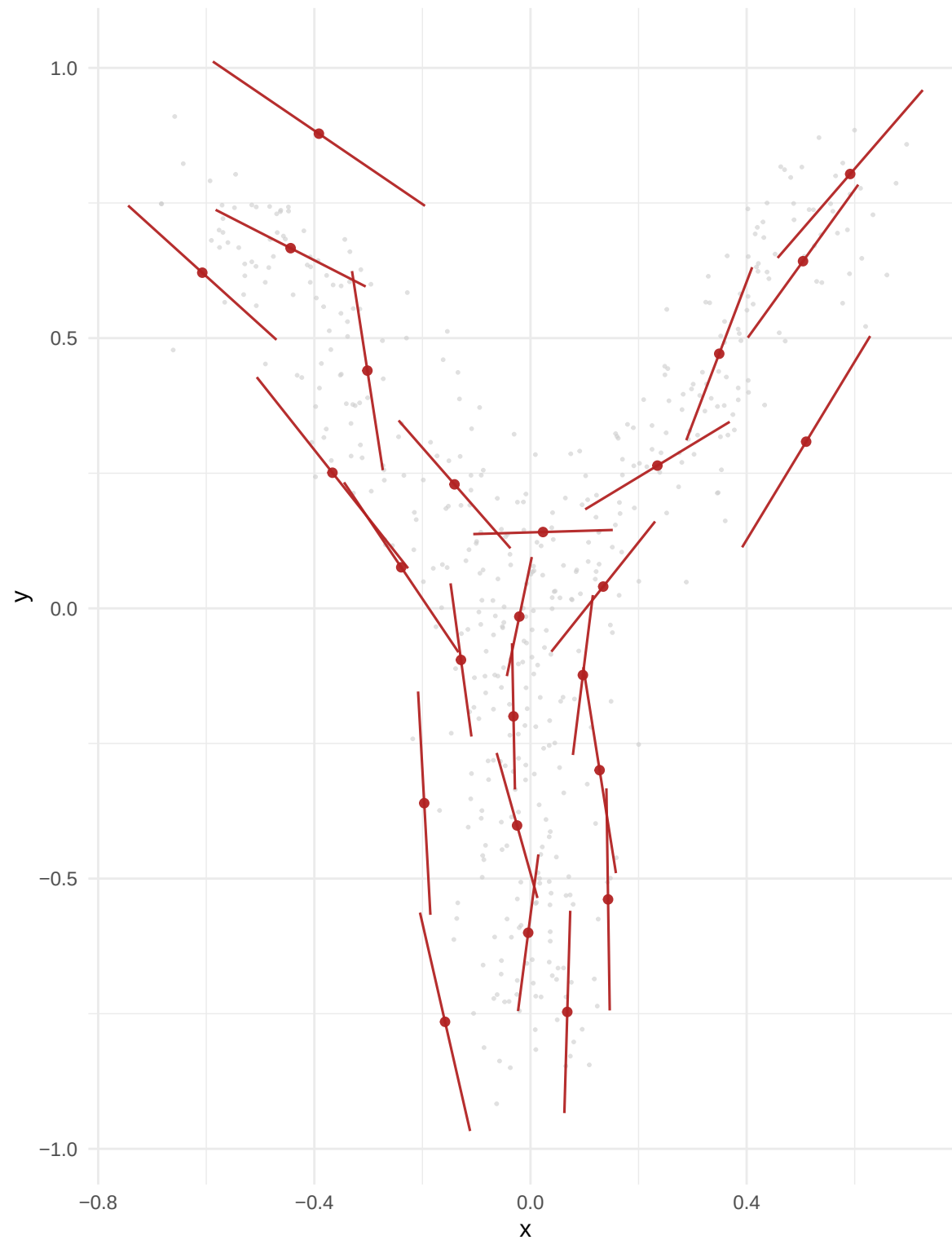


Atlas Selection (Anchors)



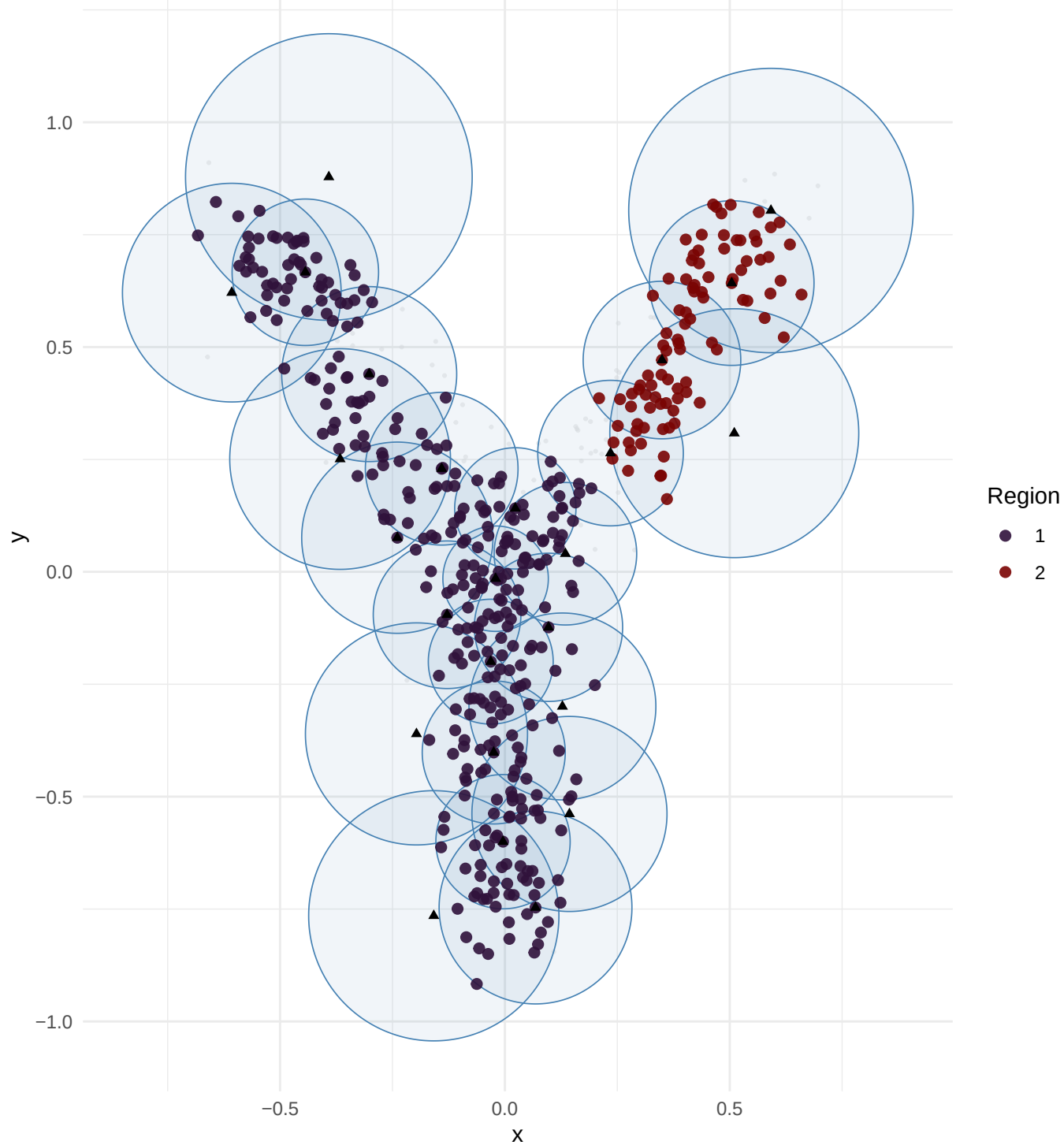
Anchor SVD Segments

Same anchors as plot 4: spheres first, then their SVD (PC1) in red



Intersection Regions (2+ anchors)

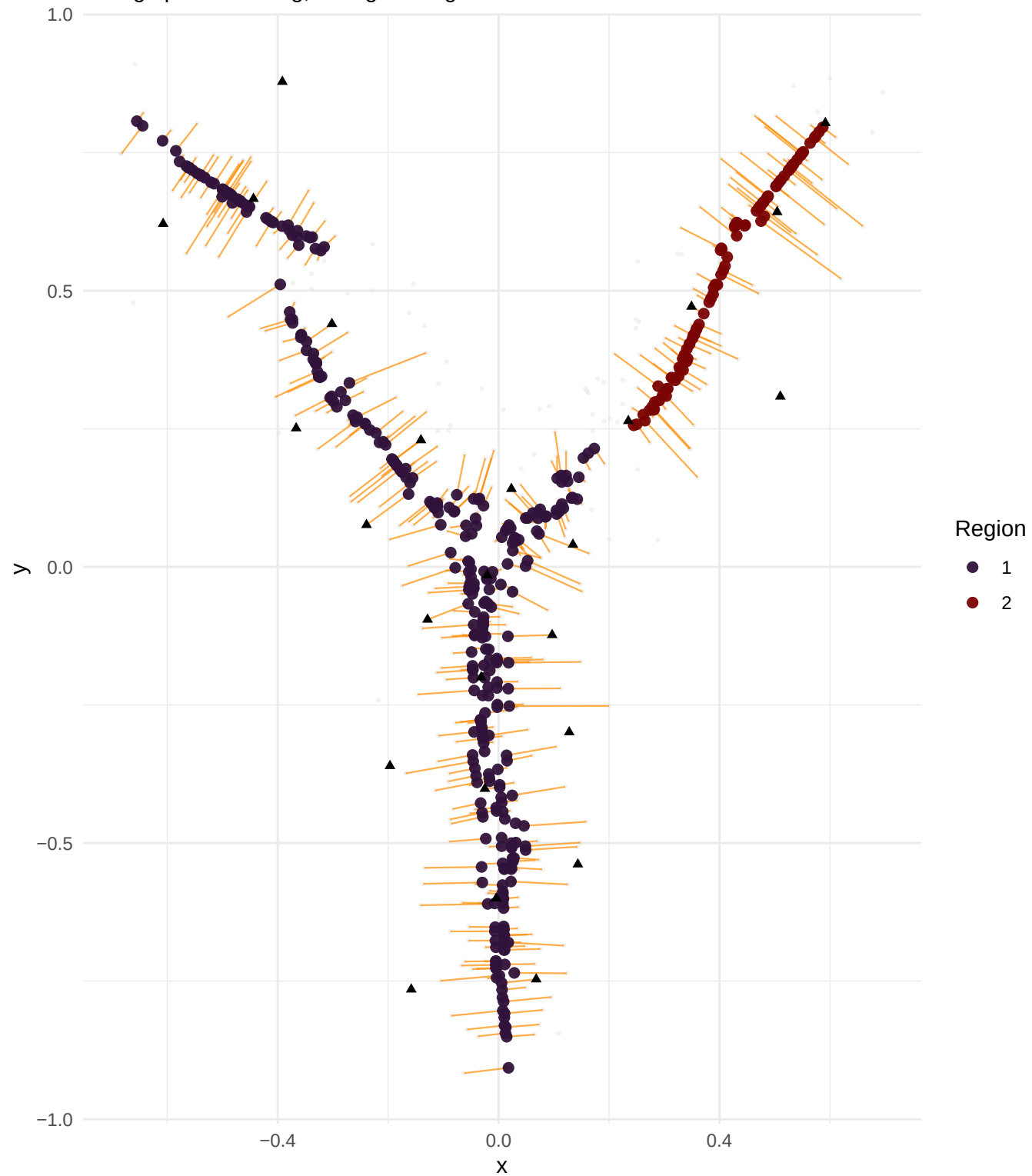
Anchor spheres in blue. Colored points = overlap zones. Triangles = anchor centers.



Parameters: k_min = 50, g_threshold = 3.0, o_max = 0.3, o_min = 0.10, stability = 0.90, scale_factor = 2.5, max_iter = 50, conv_tol = 0.01

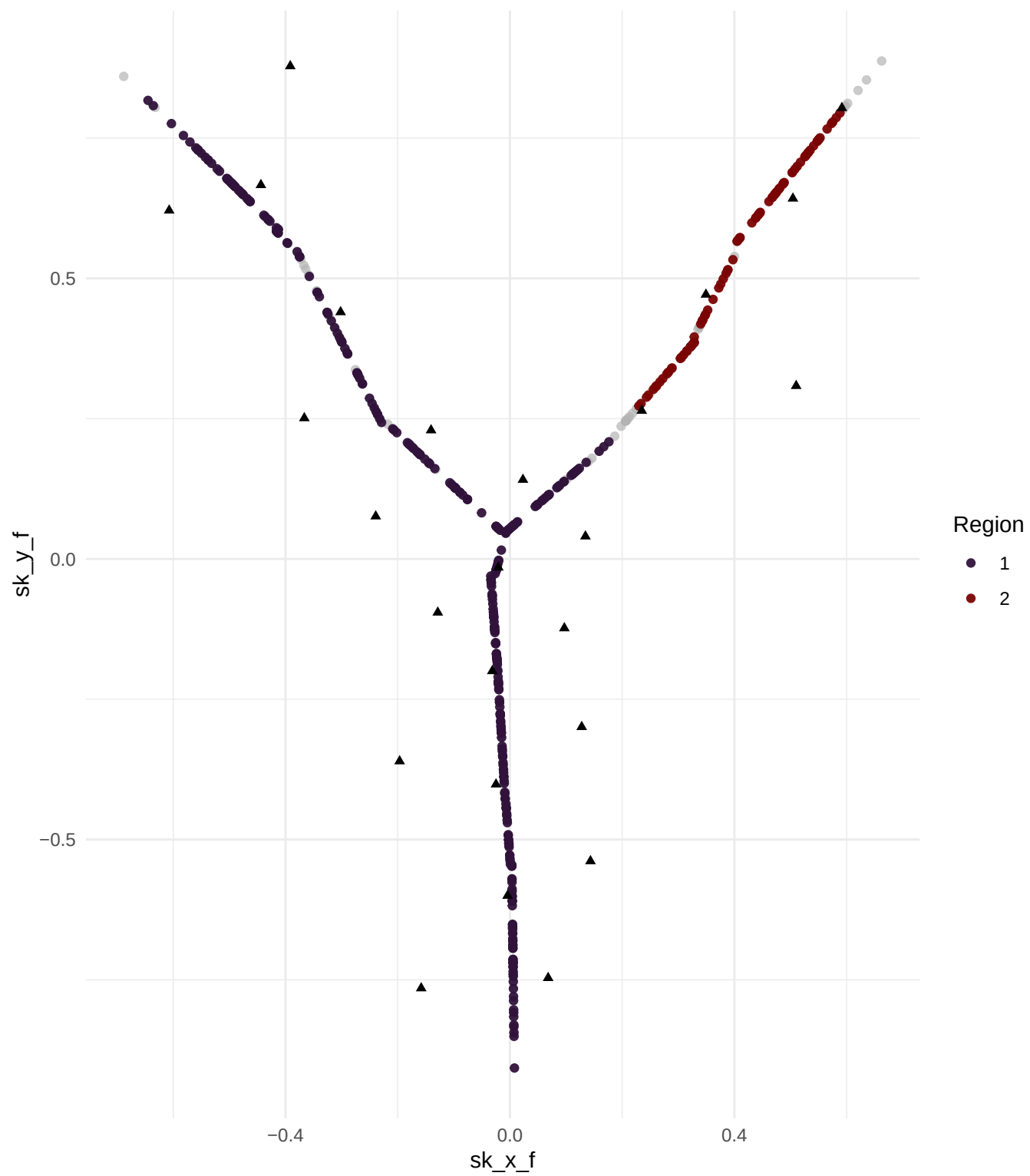
Stitched Backbone (Iteration 1)

Orange path: stitching; Triangles: original anchors



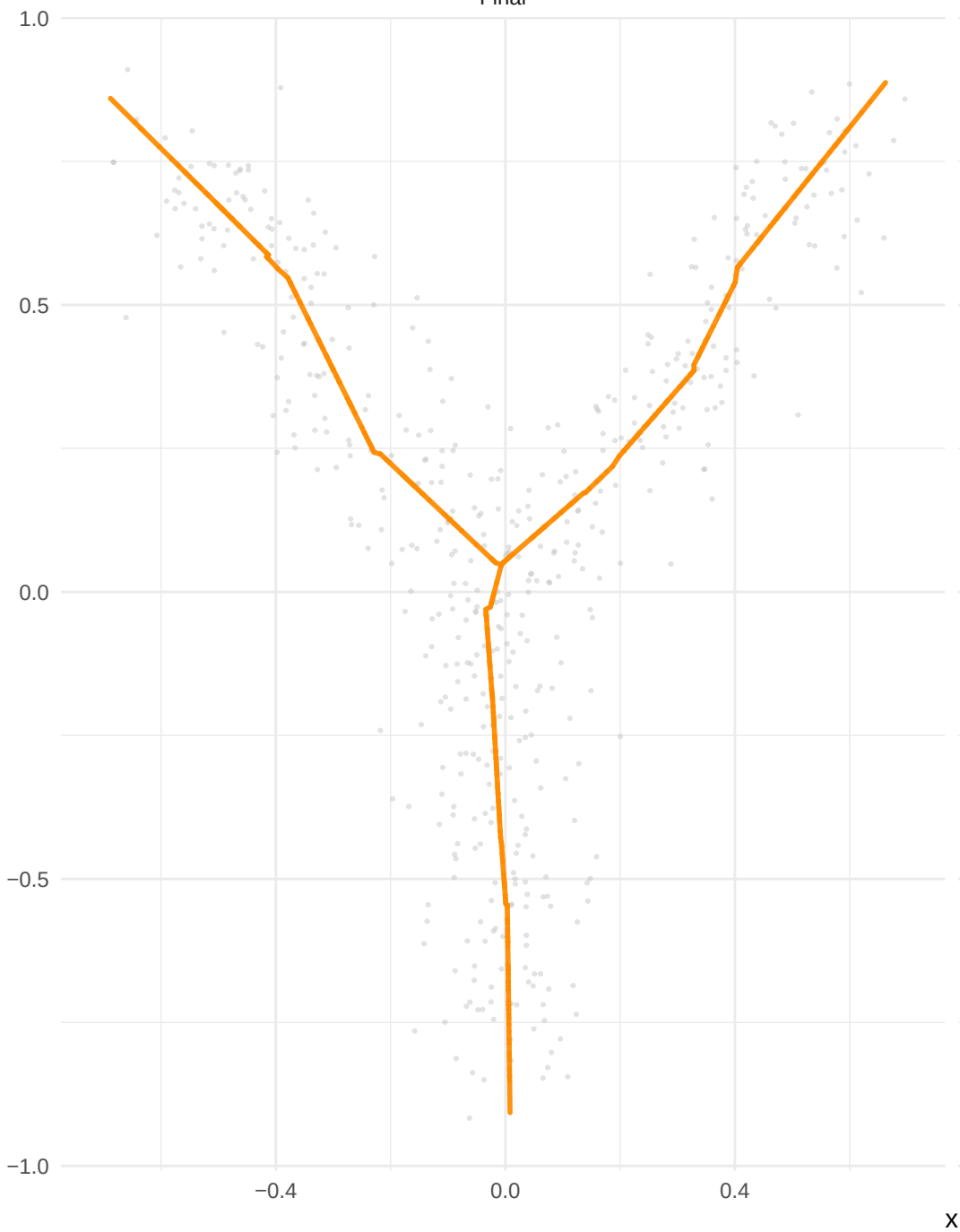
Final Stitched Backbone (Converged)

Final stitched points

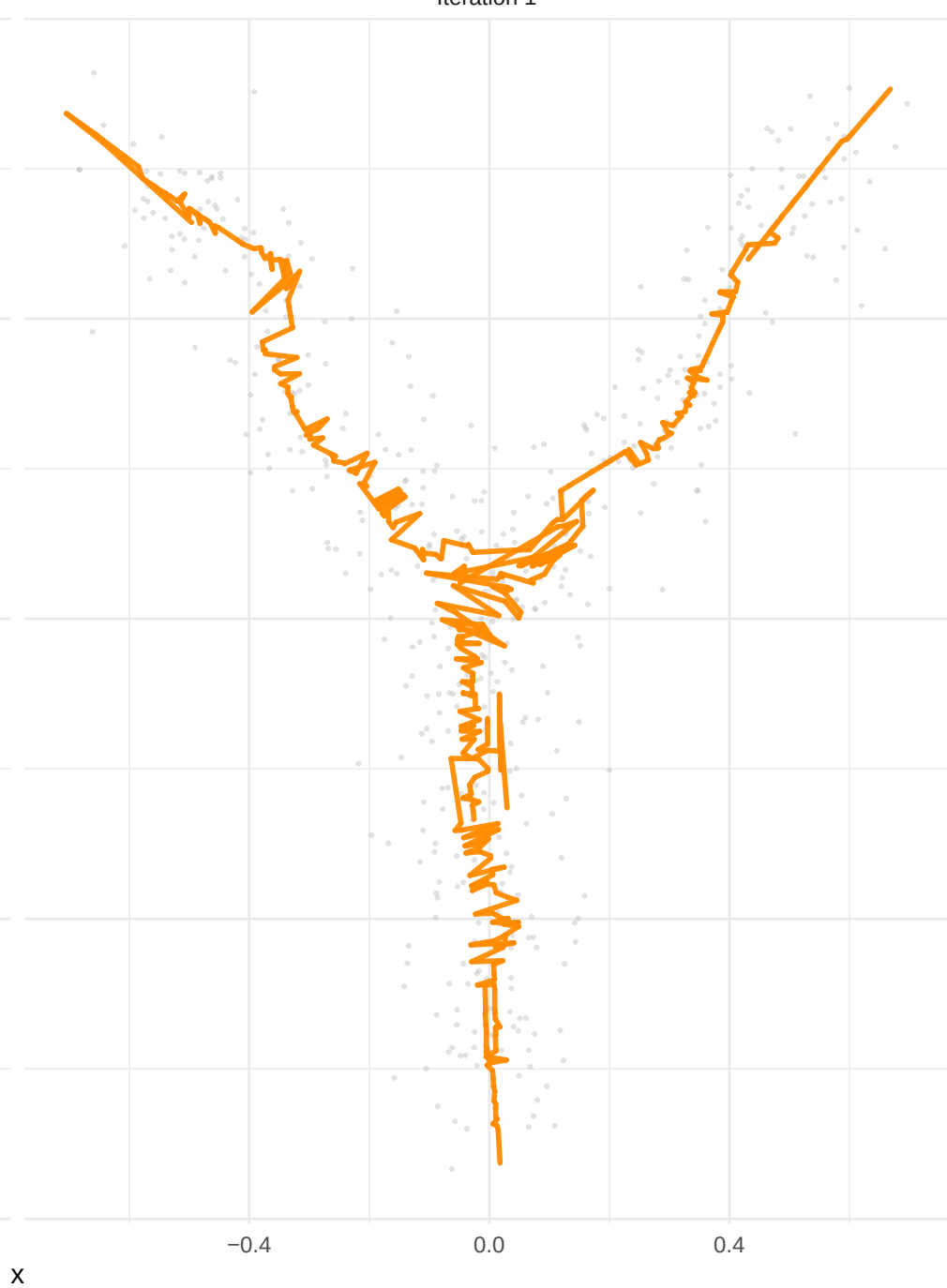


Backbone Edges

Final

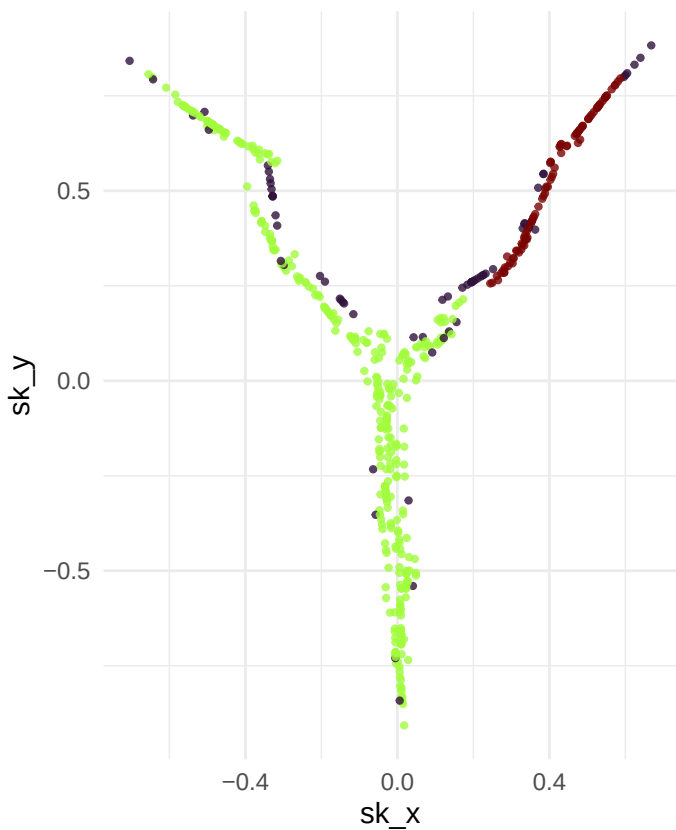


Iteration 1

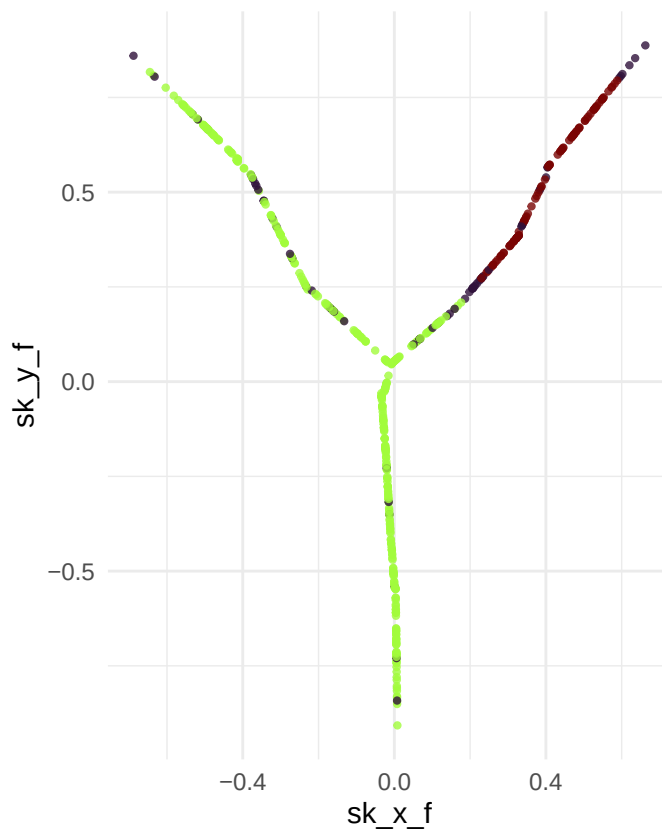


Convergence: iteration 1 vs final

Iteration 1

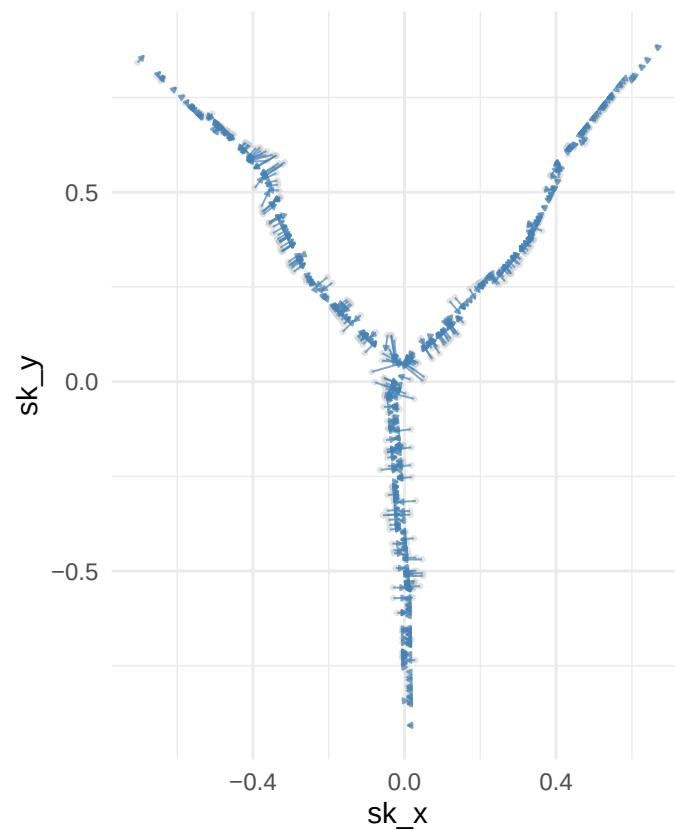


Converged



Displacement iter1 $\rightarrow$ final

Points moved $> 1e-6$: 500 / 500



Parameters: $k_min = 50$, $g_threshold = 3.0$, $o_max = 0.3$, $o_min = 0.10$, $stability = 0.90$, $scale_factor = 2.5$, $max_iter = 50$, $conv_tol = 0.01$